

RAI-818: INTELLIGENT **MOBILE ROBOTICS**

Kalman Filter – I

- **Tracking**
- **Gaussian**
- **Measuring Motion**
- **Developing a Kalman Filter**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:

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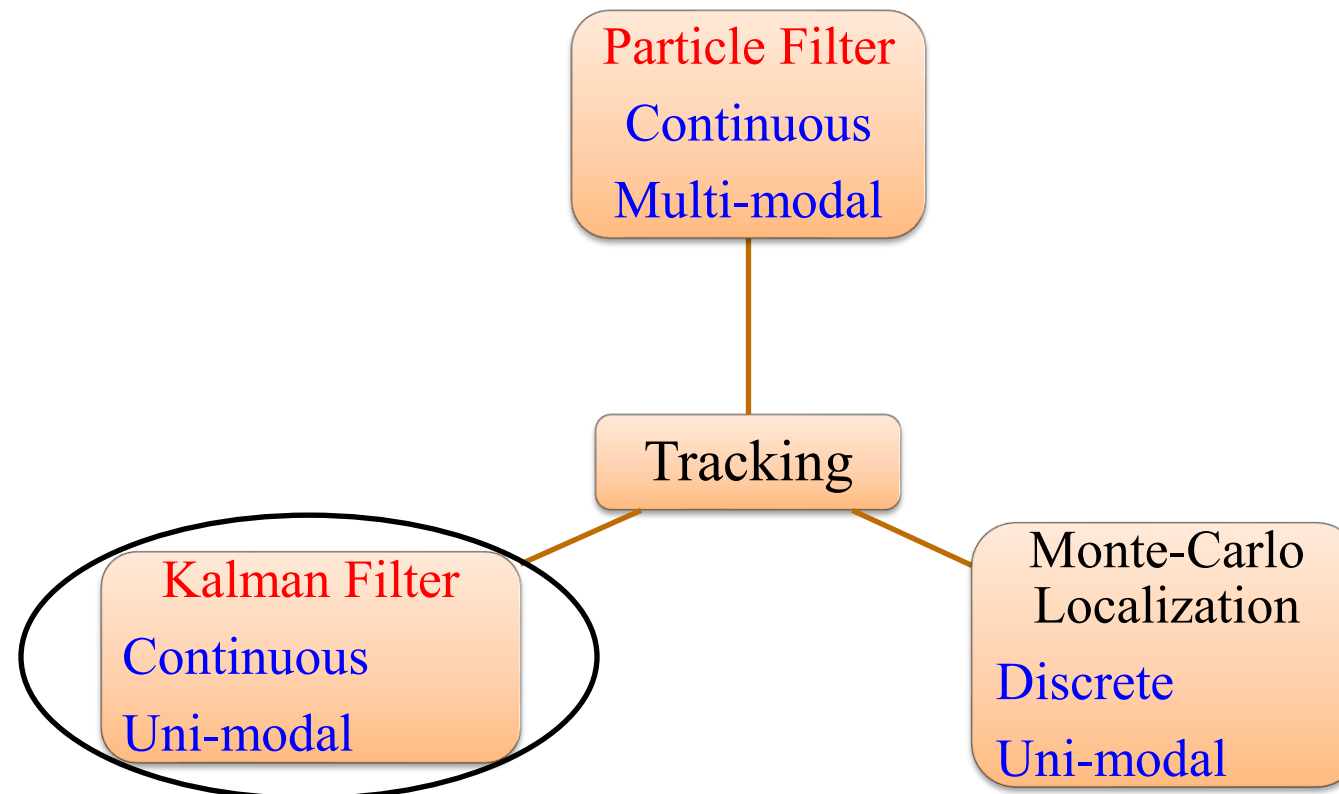
Kalman Filter



- Kalman Filter
- Tracking

The Basic problem

- The *Kalman filter* is a popular technique for estimating the state of a system. Kalman filters estimate a continuous state and gives a uni-modal distribution.?



Defining futures

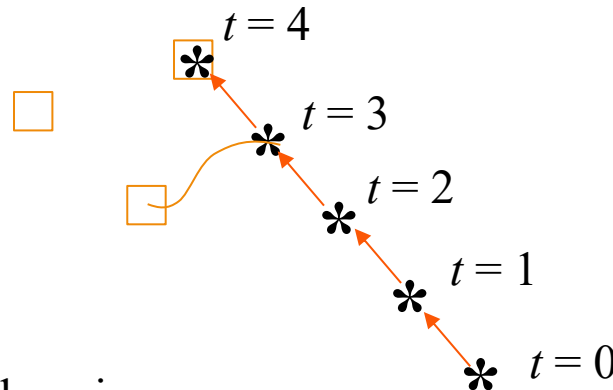
Kalman Filter



- Kalman Filter
 - Tracking
 - Example

Example

- A car senses an object in space (*) at times $t = 0, t = 1, t = 2, t = 3$. Where would you assume the object would be at $t = 4$?



- The velocity is changing according to a vector
- Assuming there is no drastic change in velocity
- We can expect $t = 4$ to be along the same vector line.

- Kalman Filter enables you take points, even if they are uncertain, like in this example, and estimate future locations
- We will also be able to identify the velocity of the object in motion.
- The Google self-driving car uses this method to understand where the traffic is based on laser and radar data.



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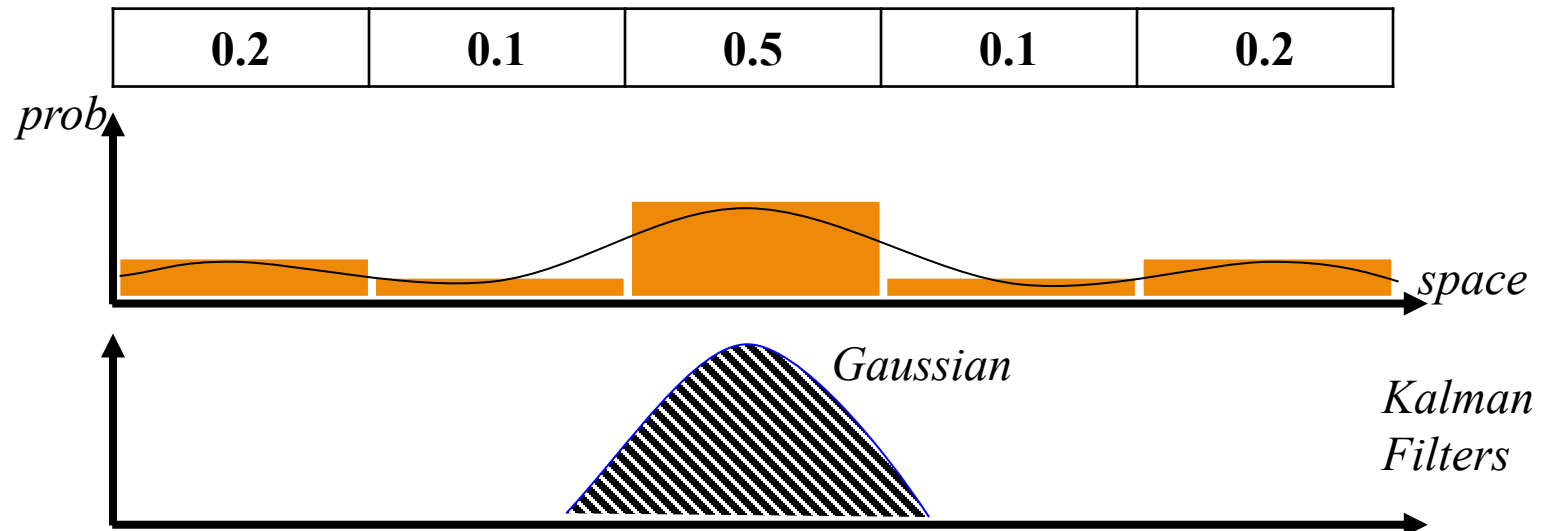
Kalman Filter



- Kalman Filter
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 - Gaussian

Gaussian

- In the previous lecture our Markov model had the world divided in discrete grids and we assigned each grid a probability
- A histogram divides continuous space into discrete regions.
- It is a way of approximating a continuous distribution.
- In Kalman filters the distribution is given in what is called a Gaussian, which is a continuous function over the space of locations. The area underneath the Gaussian adds up to one.



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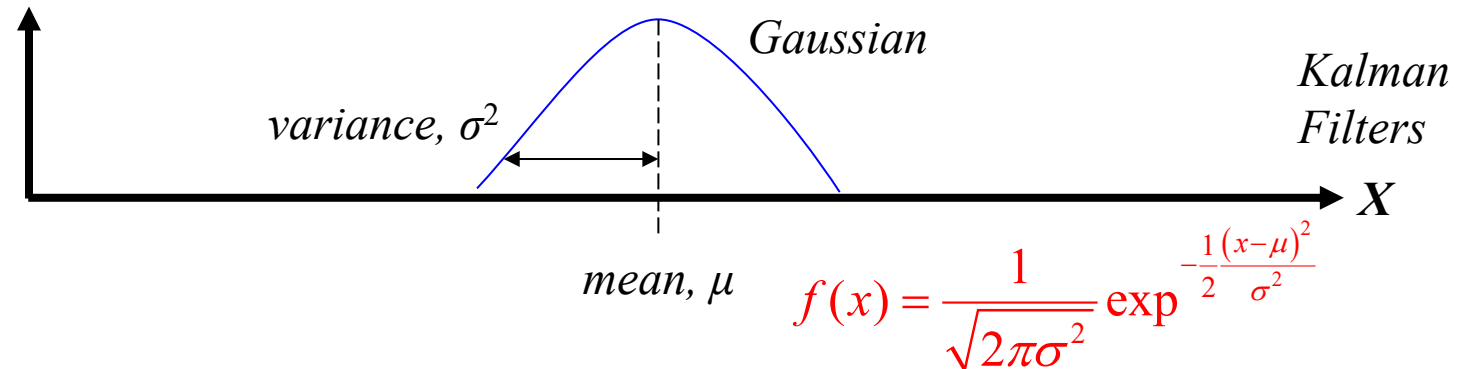
Kalman Filter



- Kalman Filter
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Gaussian

- If we call the space (x-axis), X , then the Gaussian is characterized by two parameters - the mean (Greek letter mu, μ), and the width of the Gaussian, called the variance (often written as a quadratic variable σ^2).
- Any Gaussian in a one-dimensional parameter space, is characterized by (μ, σ^2) . Our task when using Kalman filter is to maintain a μ and a σ^2 , that serves as our best estimate of the location of the objects we are trying to find.



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Kalman Filter

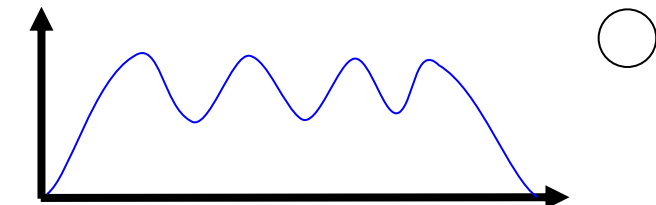
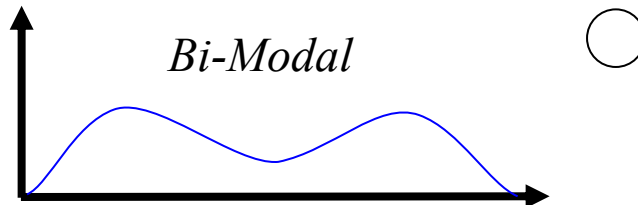
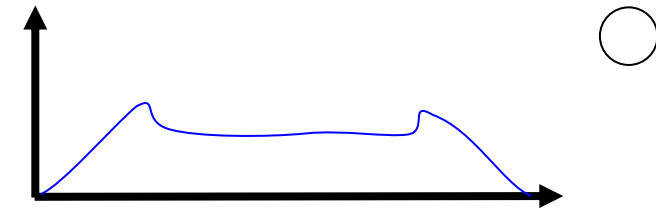
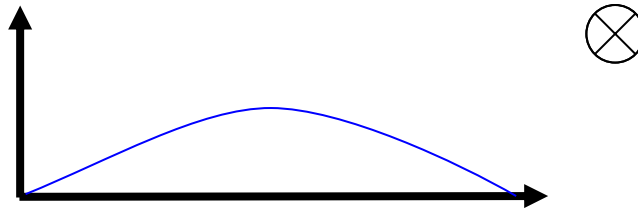
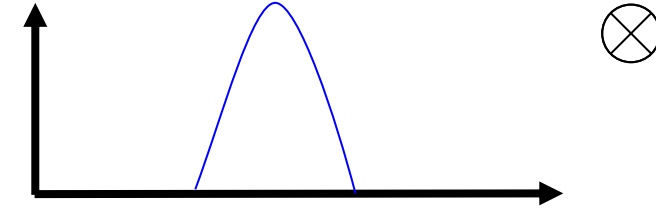
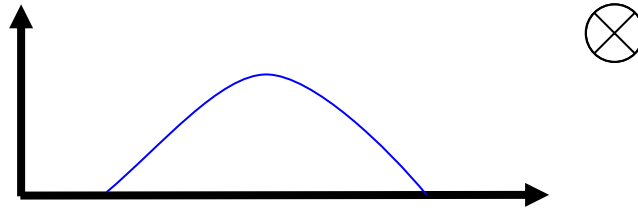


- Kalman Filter
 - Tracking
 - **Gaussian**

Gaussian

- Which of the following functions are Gaussians

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2} \right)$$



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Kalman Filter

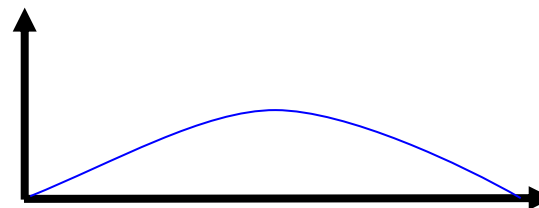
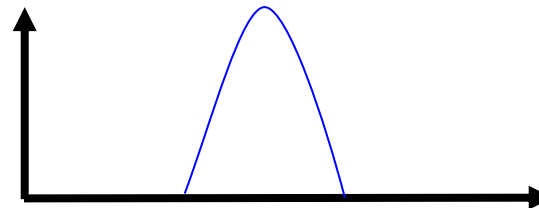
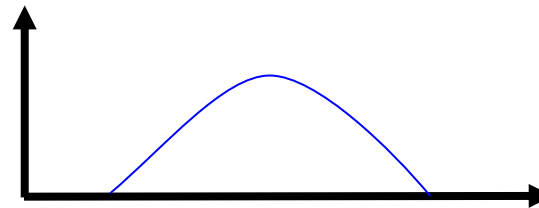


- Kalman Filter
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Gaussian

- Accurately describe the co-variance of the Gaussians.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2} \right)$$



Small

Medium

Large



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Kalman Filter



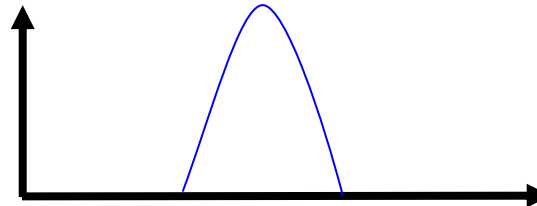
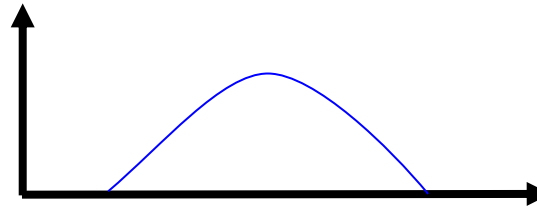
- Kalman Filter
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 - **Gaussian**

Gaussian

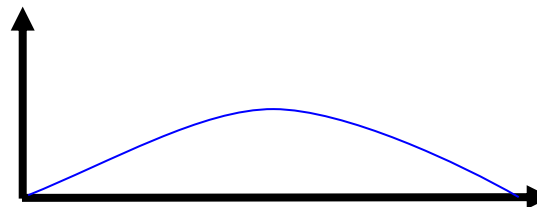
- Which Gaussian would we prefer to track a self-driving car?.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2} \right)$$

Preference



*The second is most certain,
making the chance of hitting
another car smaller*



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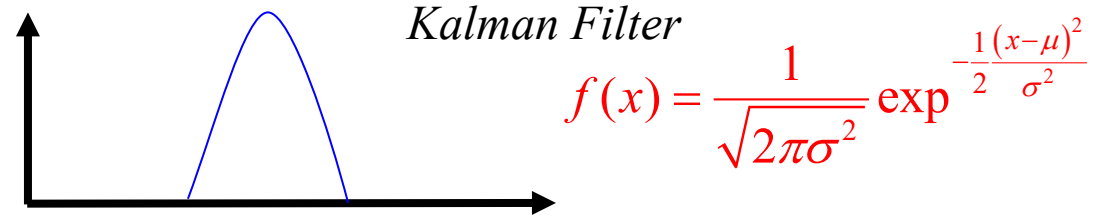
Kalman Filter



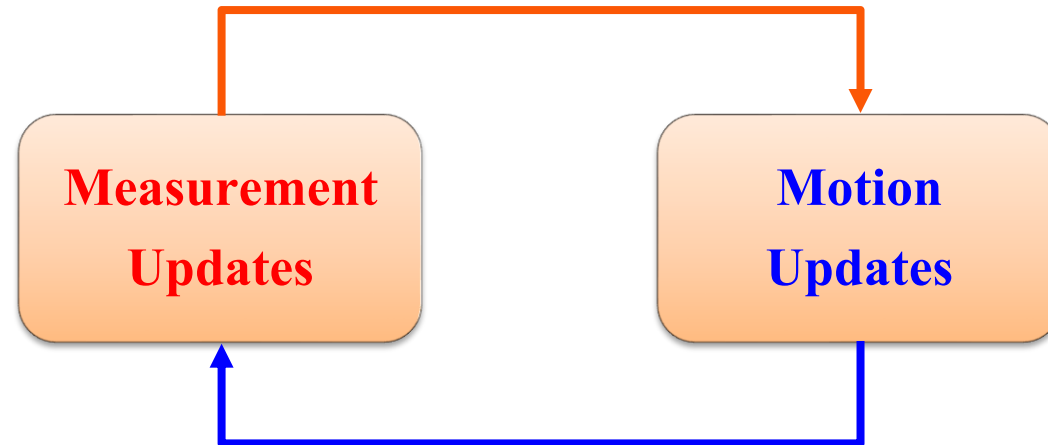
- Kalman Filter
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 - Gaussian
 - **Measuring Motion**

Measuring Motion

- The Kalman Filter represents all distributions of the Gaussians



- It iterates two things:.



Defining futures

Kalman Filter



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Measuring Motion

- Each of the two steps, measurement or motion, requires either a convolution or a product. Do you remember which requires which?

	Convolution	Product
<i>Measurements</i>	☐	☒
<i>Motions</i>	☒	☐

- Remember from previous week that when we made a measurement we multiplied our beliefs by a certain factor.
- When we made a motion we performed a convolution (which was essentially an addition of probabilities).



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Kalman Filter



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Measuring Motion

- Check whether Bayes Rule or Total Probability apply to measurements or motions?

	Bayes Rule	Total Probability
<i>Measurements</i>	☒	☐
<i>Motions</i>	☐	☒

- In previous week we used Bayes' Rule to calculate the posterior distribution given a measurement Z . We wrote this as $P(X | Z)$.
- Total probability was used in motion when we calculated the probability of being in each cell after the move by tracking the probabilities from before the move.



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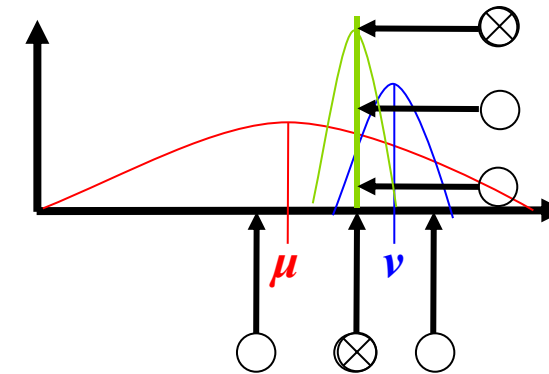
Kalman Filter



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 - **Developing a Kalman Filter**

Developing a Kalman Filter

- The measurement cycle is usually referred to as a *measurement update*, which will use *Bayes Rule* – *a product*, multiplication
- The motion cycle is usually called *prediction*. The prediction update will use *total probability* – *a convolution*, an addition
- Suppose you are localizing a vehicle and you have a prior distribution with a mean of μ as shown.
- Then we get a measurement that tells us something about the localization of the vehicle, in blue, with mean v
- Where would the new mean of a subsequent Gaussian be?
- Where is the posterior after multiplying the two Gaussians?



Defining futures

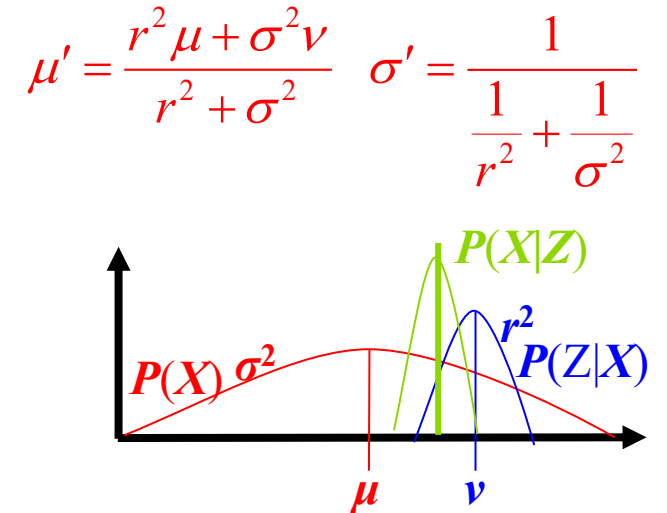
Kalman Filter



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Developing a Kalman Filter

- Suppose we multiply two Gaussians, as in Bayes' Rule, a prior μ and a probability σ^2 , and the measurement has a mean of v and a covariance of r^2
- The new mean, μ' is the weighted sum of the old means, where μ is weighted by r^2 , v is weighted by σ^2 and normalized by the sum of the weighting factors.
- The new variance term after the update is given by
- We can determine the location of the new mean because we know:
 - Prior Gaussian has a much higher uncertainty, a longer, lower slope, therefore σ^2 is larger
 - This means the v is weighted much larger than the μ
 - The mean will be closer to the v than the μ (green tick)



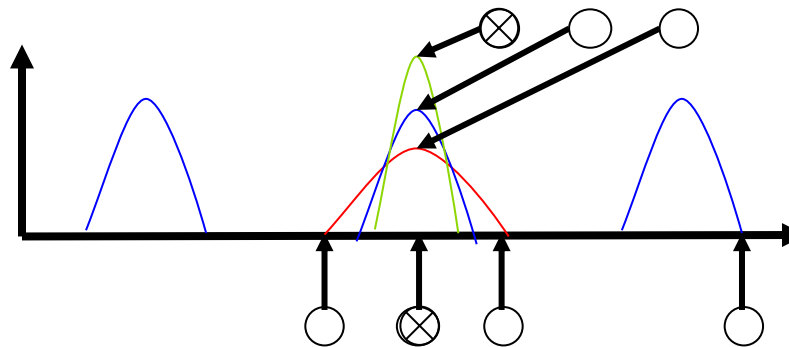
Kalman Filter



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Developing a Kalman Filter

- Suppose we have a prior and a measurement probability really far away from each other, and both have the same covariance. Where would the new mean be?
 - The new mean is in the middle because the co-variances are the same.
- What will the Gaussian look like?
 - The most peaky Gaussian, because the new σ^2 is obtained independent of the means. Again, this seems strange, but when we have two sources of information we expect our knowledge to be more precise. This explains the higher peak.



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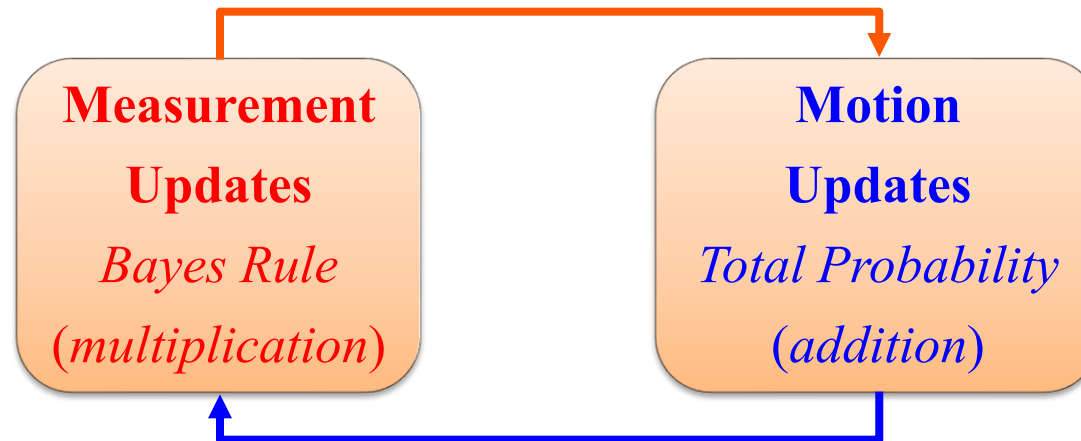
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Developing a Kalman Filter

Kalman Filter



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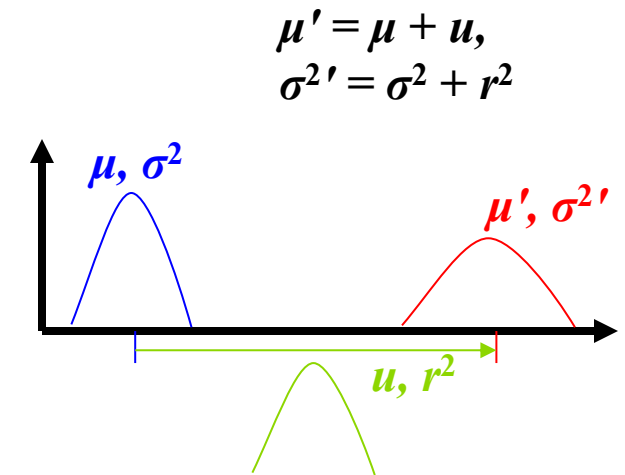
Kalman Filter



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Kalman Filter – Motion Update

- The graph below shows the best estimate of where you are (the blue tick on x-axis) and your uncertainty.
- Say you move to the right and that motion itself has its own set of uncertainty
- You then arrive at a prediction that adds the motion command to the mean and has an increased uncertainty compared to the initial uncertainty
- The New mean is the old mean plus the motion, u . The new variance is the old variance plus the variance of the motion Gaussian.
- Motion decreased our knowledge and updates increased our knowledge. This is a fundamental aspect of localization.



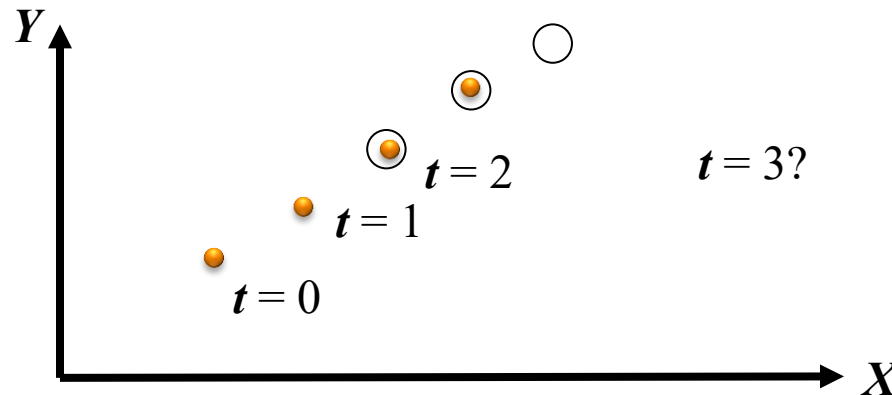
Kalman Filter



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Multi-Dimensional Kalman Filter

- Suppose you have a two dimensional state space, like a camera image. Suppose at time $t = 0$, you observe the object of interest to be at a specific coordinate.
- One time stamp later you see the object in another spot.
- Where would you expect the object to be at $t=3$? Fill in the most likely location.



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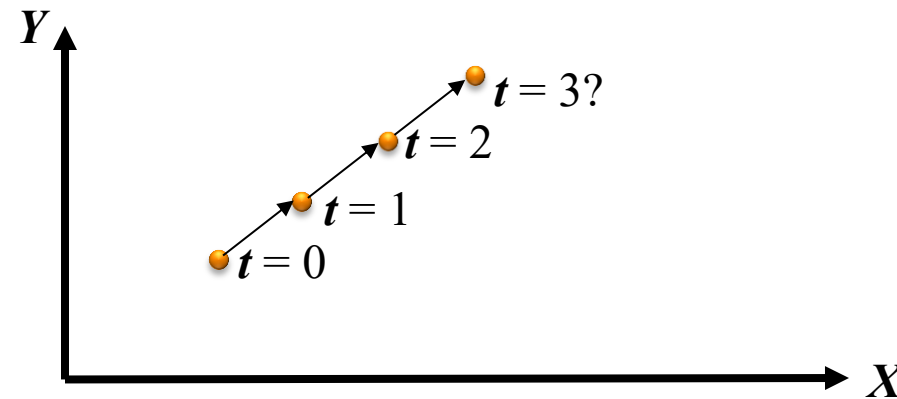
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Multi-Dimensional Kalman Filter

- In multi-dimensional spaces (e.g. real world!) the Kalman filter not only allows you to estimate your positions, but also to infer your velocity from these position measurements.
- By being able to figure out the velocity of an object the Kalman filter can make predictions about future locations that incorporate velocity.
- This is why Kalman filters are so popular in Artificial Intelligence.



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Kalman Filter



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Multivariate Gaussians

- So far we have discussed one dimensional motion, and though this captures all the essentials of a Kalman filter, we should at least briefly discuss what happens when we go from one dimension to higher dimensions.
 - Instead of being a number, the mean becomes a vector with one element for each of the dimensions.
 - The variance is replaced by the covariance, which is a matrix with d rows and d columns when the space has d dimensions.
 - This gives us a complicated formula, but it is not something you need to remember.

$$\mu = \begin{bmatrix} \mu_0 \\ \vdots \\ \mu_D \end{bmatrix} \quad \Sigma = \begin{bmatrix} \dots & \dots \\ \dots & \dots \end{bmatrix}$$



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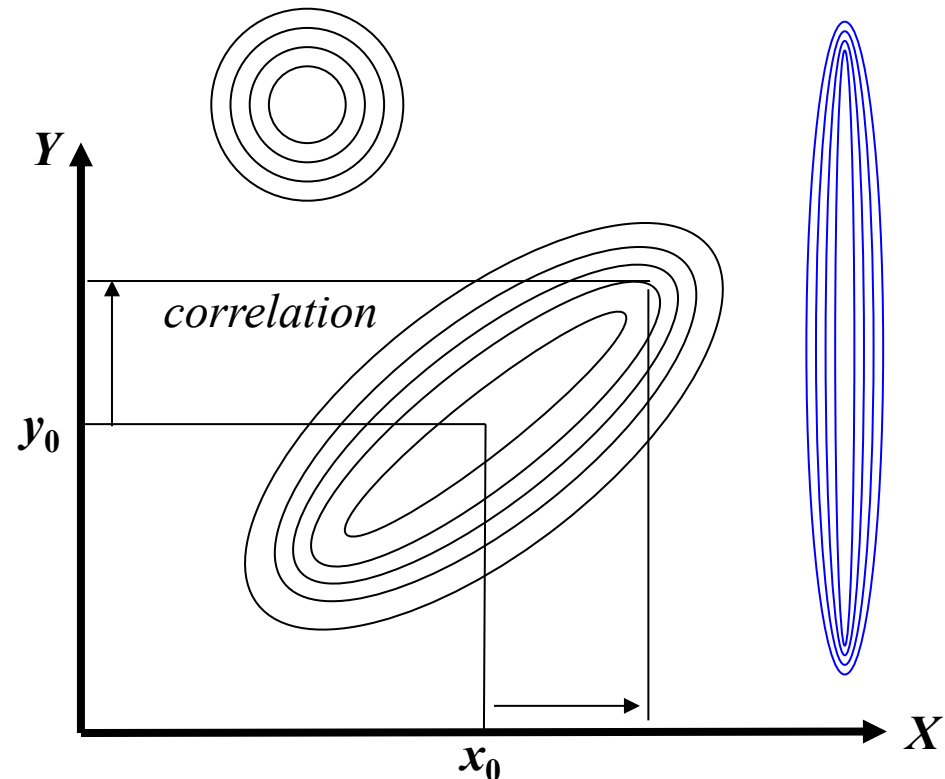
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Multivariate Gaussians

- Here is an illustration of multivariate Gaussians. In this image the mean is indicated by x_0 and y_0 and the covariance is the contour lines.
- Gaussians with a smaller amount of uncertainty would have smaller contour lines
- It is also possible to have small uncertainty in one dimension and a huge uncertainty in another
- When the Gaussian is tilted it means the x and y are correlated:



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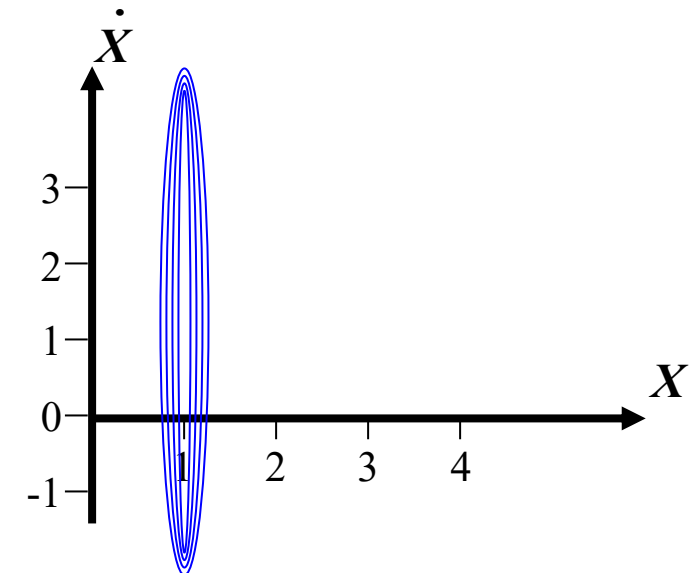
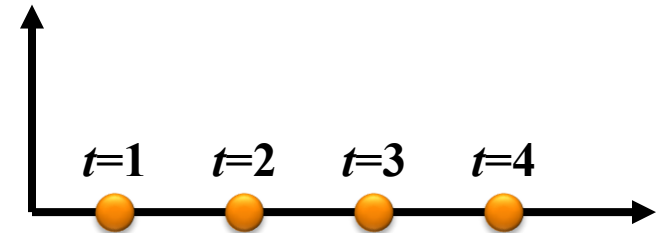
Kalman Filter



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Multivariate Gaussians

- Let us consider a 1-D motion
- Now we build a two-dimensional estimate, where the x-axis is the loc and the y-axis is the vel.
- If, initially, you know location but not velocity, you can represent it with an elongated Gaussian around the correct location, but really broad in the space of velocities.
- In the prediction step, since you do not know the velocity, you cannot predict what location you are going to assume. However, there are some interesting correlations.



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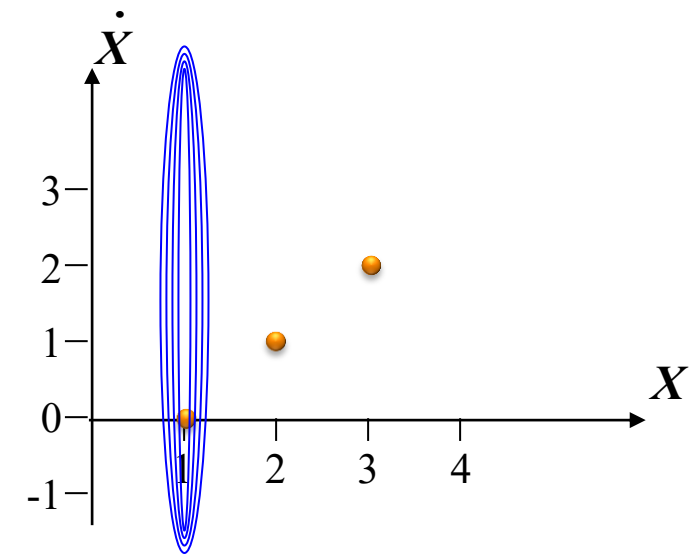
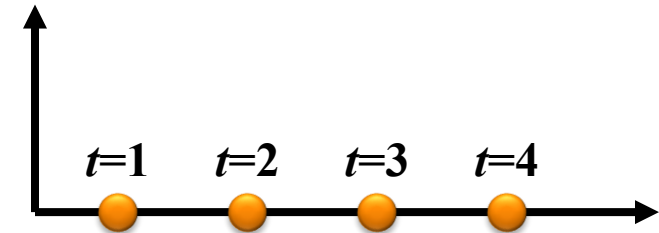
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Multivariate Gaussians

- Assume the velocity is 0, where would the posterior be after the prediction?
- If you know you start at location 1, then a velocity of zero would leave you in the exact same place, (1, 0).
- Using the same graph, assume the velocity is one, where would the prediction be one timestamp later, starting at location one (1, 0), and velocity one?
- Consider a velocity of two with the same givens as before. Where would you expect the most plausible prediction to be?



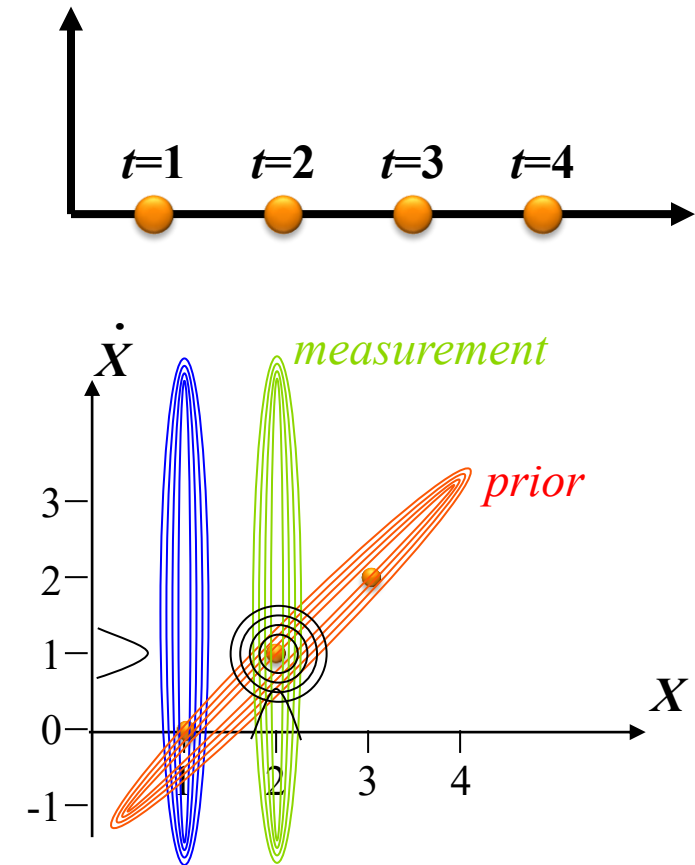
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Multivariate Gaussians

- When you find multiple data points, you find that all of the possibilities on the 1-D Gaussian (blue one), link to a 2-D Gaussian (red one).
- Consider the second observation $t=2$, which says nothing about vel, but only about the loc.
- Multiply the prior (red Gaussian) and the measurement (green Gaussian) to obtain a really good estimate of an object's velocity and location.
- When we do the math, we find that we get a peakier distribution in both the x and y directions.



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Kalman Filter

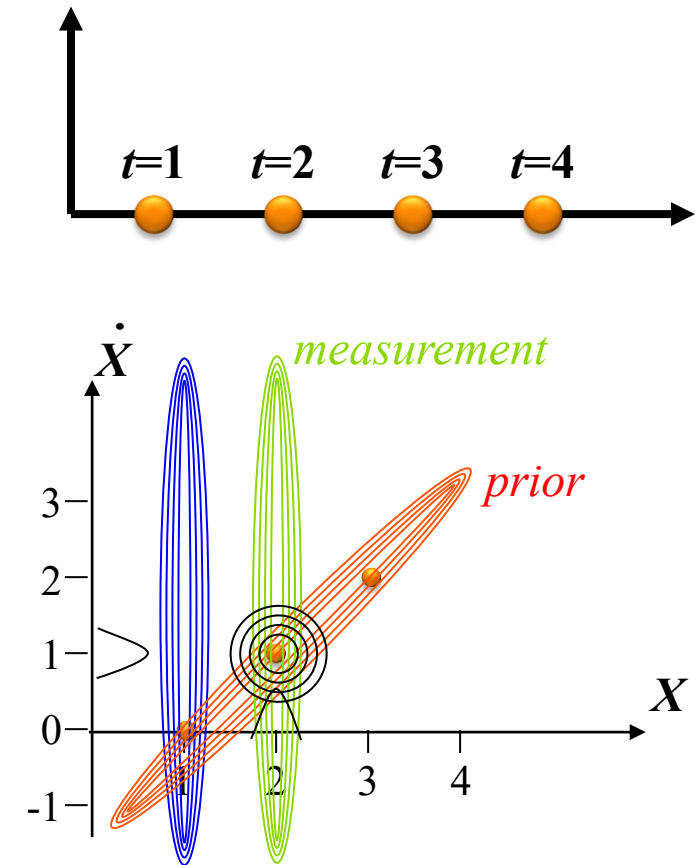


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Multivariate Gaussians

- So far we have only been able to observe one variable, and use that observation to infer another variable using a set of equations that say that the location after a time-step, is the old location plus the velocity.
- Our prediction for our new position, x' , would then just be the sum of our previous location belief, x , and our motion, u .

$$x' = x + u$$



Kalman Filter



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Multivariate Gaussians

- Variables of a Kalman filter are often called **States**, because they reflect states of the physical world, like where an object is and how fast it is moving.
- Within Kalman filters there are two sub-sets:
 - **Observables** - momentary location
 - **Hidden** – e.g., velocity. We could not measure it directly, only infer it from our position measurements.
- When these two sub-sets interact, subsequent observations of the observables give us information about the hidden variables, so that we can make an estimate of what the hidden variable are.
- From multiple observations of the places of the object's location, we can discover how fast it is moving.
- Other filters can generate this information, but the Kalman filter is the most efficient.



Defining futures

Kalman Filter

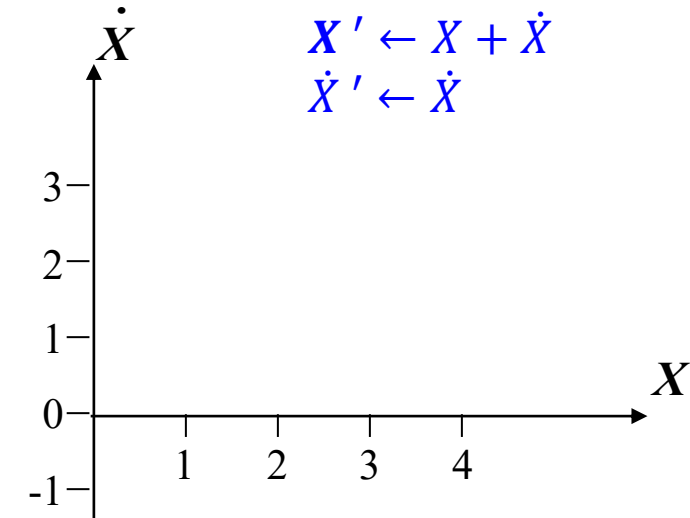


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Multivariate Gaussians

- When designing a Kalman Filter, you effectively need two things,
 - state transition function
 - measurement function,
- Both are represented as matrices. We call the state transition matrix F and the measurement matrix H .

$$\begin{bmatrix} X' \\ \dot{X}' \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}_F \begin{bmatrix} X \\ \dot{X} \end{bmatrix}$$
$$Z \leftarrow \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_H \begin{bmatrix} X \\ \dot{X} \end{bmatrix}$$



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Kalman Filter



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Update Equations

- We separate our Kalman filter into two steps: the update step and the prediction step. The equations for these steps and the names of the matrices and variables are given below.

X = Estimate

P = Uncertainty Covariance

F = State Transition Matrix

U = Motion Vector

Z = Measurement Vector

H = Measurement Function

R = Measurement Noise

I = Identity Matrix

Prediction

$$x' = Fx + u$$

$$P' = F \cdot P \cdot F^T$$

Measurement Update

$$y = z - H \cdot x$$

$$S = H \cdot P \cdot H^T + R$$

$$K = P \cdot H^T \cdot S^{-1}$$

$$x' = x + (Ky)$$

$$P' = (I - K \cdot H) \cdot P$$

- This is just a generalization of the simple one-dimensional case that we have been talking about.



Defining futures